

PONMUDI M R

ELECTRONICS & COMMUNICATION ENGINEERING STUDENT |
EMBEDDED SYSTEMS & RF/SDR ENGINEER



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SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, RF/SDR signal processing, IoT, and PCB design. Built and led award-winning hardware projects spanning LoRa disaster communication, GPS-denied satellite positioning, and custom PCB fabrication — taken from prototype to national-level competition and government research submission (DRDO/iDEX, IEEE, Ministry of Education).

TECHNICAL SKILLS

Programming: C, Python Arduino Programming
Embedded C (learning) Embedded Systems

TOOLS & PLATFORMS

ESP32, Arduino Mega RF/SDR Engineering
Raspberry Pi PCB Design & IoT

EDUCATION

B.E. Electronics and Communication Engineering

2026 - 2029

Bannari Amman Institute of Technology (BIT), Sathyamangalam, Tamil Nadu

Relevant coursework in Embedded Systems, RF/SDR Signal Processing, PCB Design, and IoT.

PROJECTS & EXPERIENCE

LoRa Disaster Communication System

National-Level Project

Designed and built a LoRa-based mesh communication network for disaster scenarios without internet infrastructure. Submitted to DRDO/iDEX and Ministry of Education research programs.

Achieved award-winning recognition at national-level hardware competitions.

GPS-Denied Satellite Positioning System

IEEE Submission

Developed an alternative positioning system operating without GPS in signal-denied environments.

Implemented RF/SDR signal processing algorithms on embedded hardware platforms.

Presented findings at IEEE-affiliated events and submitted technical documentation.

Custom PCB Design & Fabrication

Hardware Project

Designed and fabricated custom PCBs for embedded systems projects from schematic to prototype.

Integrated ESP32, Arduino Mega, and Raspberry Pi with various sensor interfaces.

Took projects from initial prototype stage through to competition-ready hardware.

LANGUAGES

Tamil, English



AWARDS/ACTIVITIES:

Award-Winning Hardware Project (National Level),
DRDO/iDEX Research Submission, IEEE Project
Submission, Ministry of Education Research Contribution